

Nassima RADOUANE

Doctor in physics and materials science (Dual Ph.D.)

Personal Details

Full name: Nassima Radouane

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Education and Diploma

📄 2017-2021: Dual PhD student

Study field: molecule and condensed matter physics.

PhD research: “Elaboration of zinc phosphate glass/graphite composites: modelling and experimental study of thermoelectric transport parameters”.

Laboratory : Unité de Dynamique et Structure des Matériaux Moléculaire EA 4476 (UDSMM).

Institution : Université Littoral Côte d’Opale (ULCO), Dunkerque, France.

Laboratory: Laboratory of composite materials, polymers and environment (LMCPE).

Institution: Mohammed V University - Faculty of Sciences-Morocco (UM5).

📄 2013-2016: metallurgy & quality control engineering

Study field: Metallurgy & Quality control.

Institution: National School of the Mineral Industry of Rabat (ENSMR).

Location: Rabat-Morocco.

📄 2011-2013: DEUG degree

Study field: Physics science.

Institution: University of Hassan I- Polydisciplinary Faculty of Khouribga.

Location: Khouribga- Morocco.

📄 2010-2011: High school degree

Study field: Mathematical sciences A.

Institution: Errazi high school.

Location: Settat-Morocco.

Professional experience

📄 End of studies project

Topic: “Evaluation of the efficiency of the burners of the ceramic oven and estimation of energy losses”

Company: Super Ceram- Kenitra- Morocco

Duration: Mars 2016/ June 2016

📄 Engineering Internship

Topic: “Study of corrosion phenomena of borosilicate glasses and stainless steel (316L and 904L) materials in phosphoric acid solutions”

Location: OCP SA- Jorf Lasfar- Morocco.

Duration: July 2015/ September 2015

First professional training

Topic: “Geotechnical testing before construction”

Location: “LCMS- Maroc”- Salé.

Duration: August 2014/ September 2014

Research experience

-  Elaboration of composite materials-based glass phosphate matrix and polymer matrix filled particles in relation with my PhD research.
-  Structural characterization: X-ray diffraction, Fourier-transform infrared spectroscopy and metallographic examination using Scanning Electron Microscopy (SEM).
-  Mechanical properties using stress-strain characterization.
-  Identification of thermal properties (Effectivity, diffusivity and conductivity) using laser infrared photothermal radiometry (PTR) technic.
-  Electrical measurement using impedance spectroscopy technic, or 4-point probes method.
-  Implementation and instrument setup for measuring Seebeck coefficient at UDSMM laboratory.

Conference and Scientific Workshop

-  Oral communication: “Thermoelectric Figure of Merit of Zinc Phosphate Glass-reinforced Graphite Composites”. Virtual Conference on Thermoelectrics, July 21-23, 2020.
-  Oral communication: “Investigation on wettability and mechanical properties of novel zinc phosphate glass-based epoxy composites”. 2020 International Symposium on Water, Ecology and Environment (ISWEE 2020), December 6-8, 2020 Online.
-  Oral communication: “Thermal and structural characterizations of epoxy/graphite composites” in JCAT50 on 03-05 January, Saint-Valéry en Caux.
-  Poster: “Structural and photothermal radiometry characterization of zinc phosphate glass-based composites” in the International Conference for Research on Phosphates and Derivatives (Phosphate days 2018) on 12-13 November, 2018 at Mohamed VI University (UM6).
-  Participation in a workshop "Simulation using COMSOL Multiphysics" on 24 mars 2017 at Ecole Normale Supérieure (ENS), Rabat.

Scientific publications

-  N. Radouane et al., “Thermoelectric improvement of the figure of merit of zinc phosphate glass composites by a likely tunnel percolation mechanism,” J. Appl. Phys., vol. 129, no. 15, p. 155110, Apr. 2021. [\[Link\]](#)
-  N. Radouane et al., “Synthesis, mechanical, thermal, and electrical characterization of graphite–epoxy composites,” J. Chinese Chem. Soc., p. jccs.202000490, Apr. 2021. [\[Link\]](#)
-  N. Radouane, M. Hammi, A. Maaroufi, B. Ouaki, M. Depriester, and A. Hadj-Sahraoui, “Investigation on Wettability and Mechanical Properties of Novel Zinc Phosphate Glass-Based Epoxy Composites,” in IOP Conference Series: Earth and Environmental Science, 2021, vol. 690, no. 1, p. 12047. [\[Link\]](#)
-  N. Radouane et al., “Thermal, electrical and structural characterization of zinc phosphate glass matrix loaded with different volume fractions of the graphite particles,” J. Non. Cryst. Solids, vol. 536, p. 119989, May 2020. [\[Link\]](#)

Software skills

↳ **Programming languages:** VBA, C++, C, MATLAB.

↳ **CAO:** CATIA, ANSYS, LABVIEW.

Languages

Arabic: Native language

French: Usual

English: Professional

Grants

↳ Grants: Researcher excellence grant provided by Moroccan National Scientific Center (CNRST: Grant no. 67UM5R2017).

↳ Grants: Researcher Ph. D. student grant by Université Littoral Côte d'Opale (ULCO) (Grant no. 17006508).